Evaporation

Group #1

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Introduction

The purpose of this experiment was to:

- Concentrate a solution which consists of a non-volatile solution and a volatile solution using the apparatus.
- See how concentration, capacity and economy are affected by operating pressure.

Theory

- Evaporation is the phase change from liquid to vapor.
    - It is a heat transfer operation used as a separation process by utilizing volatility differences using a steam fed heat exchanger and a condenser.

- The following equations are used to produce conclusion:

- **Heat Transfer Equation:**

  $\boldsymbol{q = UA\Delta T}$, where:

  q: the rate of heat transfer (J/s)

  U: the overall heat transfer coefficient (W/(m^2*k))

  A: the area available for the heat transfer (m^2)

  ΔT: the average temperature difference for heat transfer (oC)

- **Capacity:** $\boldsymbol{\frac{Vd * \rho w}{t} = \frac{Vdt}{t}}$, where:

  Vd: Volume of distillate (mL)

  ρw: Density of water (kg/L)

  t: Time (hour)

- **Economy:** $\boldsymbol{\frac{Mass\ of\ product\ (kg)}{Mass\ of\ steam\ used\ (kg)}}$

Apparatus

#	Name	Description
1	Water Makeup Tank	Holds water
2	Condenser	Convert evaporated vapor to liquid
3	Rotameter	Measures the flow rate of condensed liquid
4	Pressure Gauge	Measures pressure of system
5	Distillate Collection Flask	Collects the distillate form the vessel
6	Vacuum Pump	Pressurizes the system
7	Vapor Liquid Separator	Holds the Sugar mixture for evaporating
8	Power Pump	Switch for power
9	Heat Exchanger	Heats makeup water using steam
10	Steam Trap	Controls steam
11	Steam Collection Flask	Collects resulting steam from vessel
12	Digital Thermometer	Measures Temperature at steady state
13	Steam Valve	Opens and closes steam flow

Apparatus

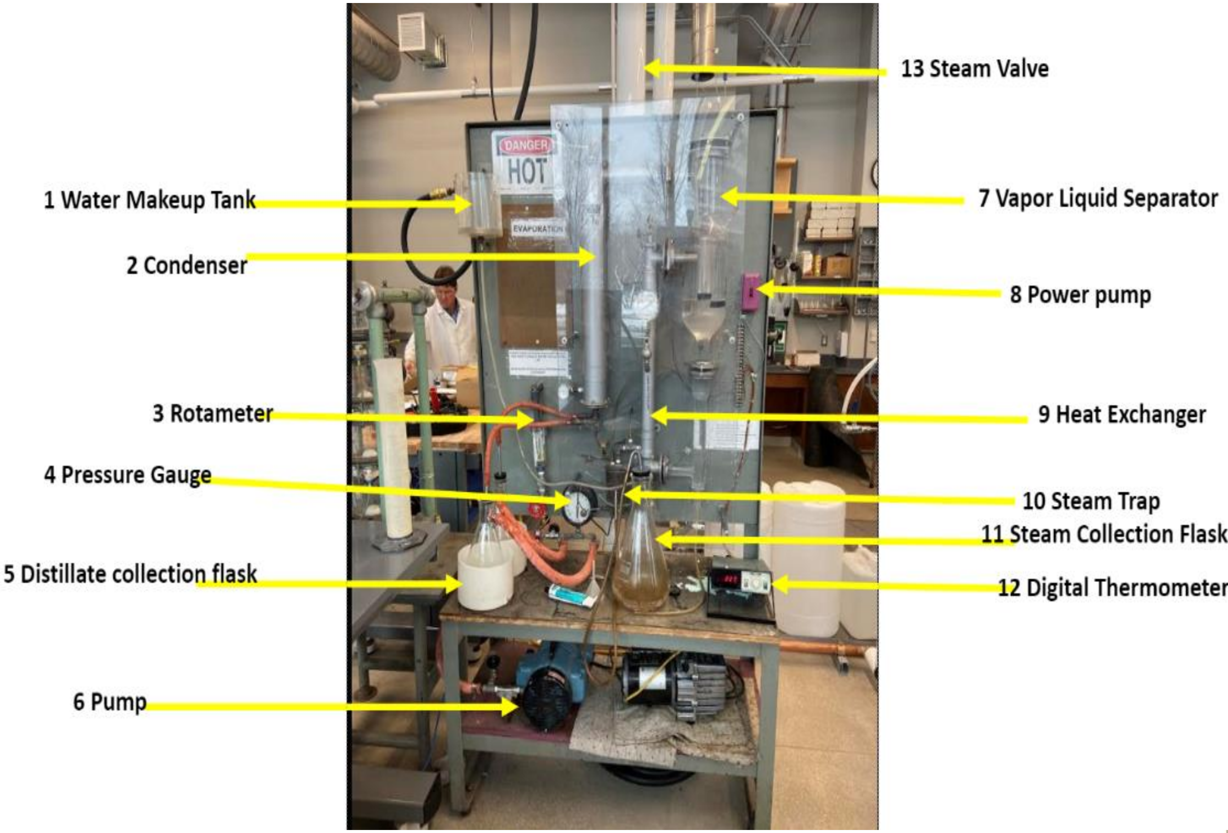

Figure 1: Main Apparatus

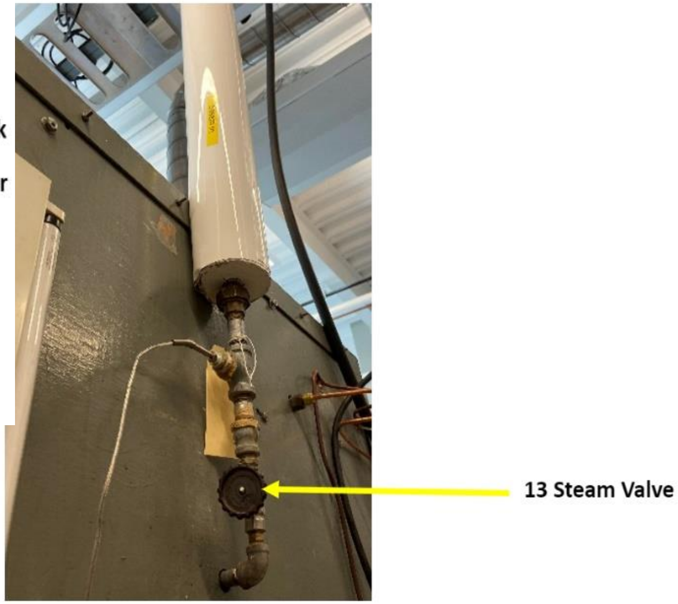

Figure 2: Steam Valve

Material & Supplies

Materials/supplies	Usage
Cooling Water	Used in the condenser
Electricity	Used to run the pump
Sugar and water	Used to mix the sugar solution
Heat Resistant Gloves	Safety precaution to protect from burns
Nitrile Gloves	Safety precaution to protect from skin contact
Steam	Used in the heat exchanger

Experiment Procedure

1. Drain contents of the separator through a valve at the bottom of the separator.
2. Prepare a 10 wt% sugar solution and fill the separator to the level line. Allow some of the solution to flow into the tube used to drain the separator.
3. Put pure water in the makeup tank- adjust the flow of makeup water until it is at a constant level. Start the flow of steam into the system.
4. Start the cooling water to the steam trap. Collect the steam condensate in a graduated cylinder.
5. Turn on the vacuum pump and adjust the bypass valve to obtain a desired pressure in the separator.
6. Adjust the vacuum so that a steady flow is achieved.

Experiment Procedure (Cont.)

7. Collect the vapor product (after it has condensed). Be sure to add enough makeup water to maintain the level at a steady state.
8. Wait for the system to reach a steady state as indicated by the temperature readings for dial setting 1 and 2.
9. After reaching a steady state, note the level of the steam condensate and collected vapor. Now, wait for 15 minutes and read these levels again.
10. Record the flow rate of the cooling water to the steam trap, and the inlet and outlet temperature.
11. Now, change the pump to a different vacuum level, for a total of three vacuum settings

Experimental Challenges

- Timing with precision
- Check if there is water in the apparatus before starting
- Functionality of equipment

Safety

- Be careful with electrical/pump equipment.
- Do not get water on the exposed wires
- Wear heat gloves
- Be careful of any spills.